Magnitude and Flux Density of RU Camelopardalis

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Abstract

RU Camelopardalis, also known as RU Cam, is a Type II Cepheid variable star. We determine that the magnitude and flux density of RU Cam in the V-band are 10.8 and 1.71×10^{27} W m^{-2} Hz^{-1}, respectively. To find these values, I imaged RU Cam using a 14" telescope while also taking calibration measurements and attempting to image a standard star. Using Python, SAO DS9, and astrophotometry, I analyzed the images and determined the aforementioned values.

1 Introduction

RU Camelopardalis, otherwise known as RU Cam, is a Type II Cepheid variable star of the W Virginis variable subgroup in the constellation Camelopardalis.[4] It has an average period of 22 days, Kollath states.[4] According to the SIMBAD astronomical database, RU Cam has a right ascension of $07^h21^m44^s$ and a declination of $+69^o$ 40' 15". [8] Kipper states RU Cam is a carbon star. [3] Its atmosphere contains more carbon than it does oxygen. Kipper [3] notes that there are not many Type II Cepheid stars that are also carbon stars, making RU Cam an unusual case. RU Cam was discovered in 1907 by Ceraski [1] and was subsequently reported to be a new variable star with its distinction as a Type II Cepheid following soon after.

Type II Cepheid variables are stars with varying periods between one and 50 days. Variable stars are a type of star whose apparent magnitude – brightness seen from Earth – changes over a given period of time, according to Wallerstein. [7] Type II Cepheids are typically found in older stellar populations with a low metallicity, Das states. [2] RU Cam is a W Viriginis variable. [4] Type II Cepheids are commonly divided into three subclasses depending on their periods: BL Herculis variables (1-5 days), W Virginis variables (10-20 days), and RV Tauri variables (periods greater than 20 days). [7] While RU Cam exceeds the typical period constraints of a W Virginis variable, it is commonly referred to as such in publications.

Cepheid variables are useful as distance indicators, or standard candles, because of their period-luminosity relationship. [2] The period-relationship describes the directly proportional correlation between the luminosity of variable stars with their period [5]. In her 1912 paper, Leavitt[6] found that there was a direct relationship between magnitude and the logarithm of the period. Furthermore, this relationship was defined to include that the apparent magnitude of a star is logarithmically proportional to the absolute magnitude divided by a distance. [6] This relationship yields what we now call the distance modulus formula.

$$m - M = 5 \log_{10}(d) - 5 \tag{1}$$

The period-luminosity relationship is used for cepheids in distant galaxies. Astronomers calculate the distance to that galaxy using the cepheid's observed period, an absolute magnitude can be calculated. Comparing this to the apparent magnitude, one can estimate the distance to the cepheid and therefore the galaxy using the distance modulus formula.

The flux density refers to the amount of light an observer at some distance receives at given wavelength range.

$$f_\lambda = \frac{L_\lambda}{4\pi r^2} \tag{2}$$

When determining the apparent magnitude of a star in reference to another star, the stellar magnitude equation ,or Pogson's equation, is useful in conjunction with the ratio of flux densities.

$$m_1 - m_2 = -2.5 \log_{10}(\frac{f_1}{f_2}) \tag{3}$$

Where luminosity is measured within a range of wavelengths. flux density is measured in the V-band, an optical filter measuring the visual light range of wavelengths.

In this lab report, I measure the apparent magnitude and flux density of RU Cam. To do so, I compared it with a known standard star, Zeta Cassiopeiae. Zeta Cassiopeiae, or Zet Cas, is a variable star in the constellation of Cassiopeia, according to the SIMBAD astronomical database. [8]

2 Methods

I obtained data using the Boston University Department of Astronomy's 14" telescope in the Judson B. Coit Observatory. The telescope is a Celestron Schmidt-Cassegrain catadioptric model with a primary diameter of 14.25". Along with the telescope, I used MaximDL imaging software to capture and save .FITS files of stellar images and calibration frames. Additionally, the telescope uses TheSky software to remotely operate and point the telescope from a control room. This assisted us when finding and centering the telescope on RU Cam.

I visited the facilities on the nights of February 4, February 9, and February 11, 2026 to observe RU Cam and take the following calibration images: flat frames, bias frames, and dark frames. Each calibration image refined the image to show the stars more clearly by removing the imperfections. To take a bias frame, we put a lens cap on the telescope and set the exposure time to "bias" which was as short as possible in MaximDL. To take a flat frame, we positioned the telescope to focus on a white poster board inside of the observatory dome. We held a light to the poster board, illuminating the white page until the image was half-saturated. We changed the voltage of the lightbulb until the image was half-saturated in MaximDL. We took three sets of bias frames and flat frames, setting the telescope filters to those respective modes. We also took three sets of dark frames but they were ultimately not used in my analysis. In Python, I averaged together the three images of each type of calibration to create a master bias .FITS file and a master flat .FITS file.

I was not able to image a standard star on the three nights I visited the telescope due to unfavorable weather on the second and third nights. To account for this, I borrowed images from two of my classmates of the standard star Zet Cas. I averaged together three images of Zet Cas at .01 second exposure in Python to create a master standard star image, also saved to a separate .FITS file. I did the same for images of RU CAM imaged with a 4 second exposure.

I used Python to analyze and refine the image files as well as SAO DS9 software. I researched the characteristics, right ascension, and declination of RU Cam in the SIMBAD astronomical database. [8]

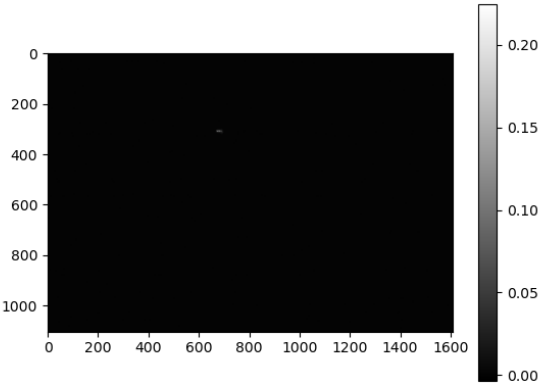


Figure 1: Corrected image of Zeta Cassiopeiae

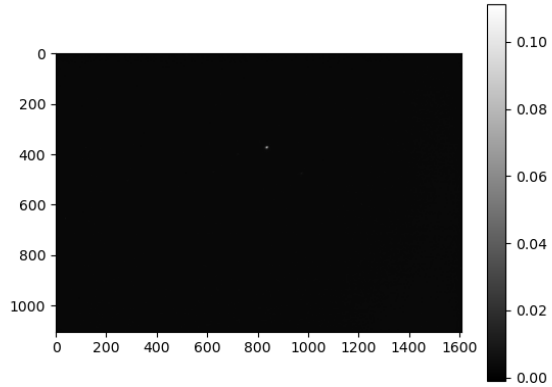


Figure 2: Corrected image of RU Camelopardalis

3 Analysis

The goal of this report and the following written code is to determine the magnitude and flux density of RU Camelopardalis using the images I took with the 14" telescope.

In a Jupyter Notebook file, I installed the photutils package, a Python library useful for astrophotometry. I also imported other useful libraries such as numpy, astropy, and matplotlib. I compiled all of the data into a folder where each .FITS file was labeled with its exposure, target (e.g. standard star, RU Cam, etc.), and/or calibration frame type. After each image was read in and averaged using the aforementioned method, I corrected the RU CAM image. To do this, I subtracted the bias frame from the main star image and divided that by a master flat frame that also had the master bias subtracted from it.[1]

This data was saved to a separate .FITS file and was imported into SAO DS9 to affirm that the image was clean and the star was clearly visible. I repeated this process for the standard star image, substituting the RU Cam image for a master standard star image.

From here, I used SAO DS9 to find the pixel coordinates for RU Cam and Zet Cas which are

$$(837, 373)$$

and

$$(681, 309)$$

, respectively. In the next line, I used these positions to write an approximation of the circular aperture and annulus – to image and subtract background noise. Each was merged with the star images and normalized, yielding the "counts," or electrons detected by a pixel in a charge-coupled device (CCD). The standard star, Zet Cas, provided a baseline to compare to RU Cam for subsequent calculations. Using ESO data, I determined that the apparent magnitude of Zet Cas in the V-band is 3.66. Using the stellar magnitude equation, I calculated the magnitude of RU Cam with respect to the ratio of counts per second, correlating to the exposure time.[2]

The zero point flux density of the V-band is $3.64 \times 10^{-23} W m^{-2} Hz^{-1}$ I calculated the flux density of the standard star using the following relationship:

[1] See attached code

[2] See attached code

$$m = -2.5 \log_{10}(\frac{f}{f_0}) \quad (4)$$

After finding the flux density of Zet Cas, I calculated the flux density of RU Cam with respect to the ratio of counts per second for the stars. [3]

4 Discussion

Results

The apparent magnitude of RU Cam according to the images taken on February 4, 2026 at 10:27 p.m. is 10.8. The accepted magnitude of RU Camelopardalis is between 8.5 and 9.5 in the V-band, according to Zaitseva. [9] Given this, the measurement procured in this report has a 20% error. The flux density of RU Cam is 1.71×10^{27} W m^{-2} Hz^{-1} according to images taken on that date at that time. There is no flux density that I could find in literature to compare the calculated value.

The average value of the three bias exposure frames is 240.8 Analog-to-Digital Units (ADUs). The average value of the three dark exposure frames is 241 ADUs. The 14" telescope has a 13-bit resolution, therefore its saturation limit is 8,192 ADUs.

Astronomers measure the flux density of a star to determine the amount light an observer receives at a given range of wavelengths. In this case, the V-band for visible light is appropriate. Astronomers measure the apparent magnitude, brightness seen at a given location, to determine a star's luminosity with respect to its distance from the observer. I believe it is worthwhile to note that RU Cam is a variable star at an unknown point in its period. Its magnitude fluctuates between 8.5 and 9.5 over 22 days. [9]

[3] See attached code

Explanation of Error

The significant margin of error could be attributed to multiple causes. One such cause is the presence of Earth's atmosphere. The 14" telescope is a ground-based telescope, therefore Earth's atmosphere could blur the image. Light is distorted by atmospheric particles which makes the star look blurry, making it difficult to pinpoint its exact pixel location in SAO DS9. Additionally, these images were taken at night in Boston, a large city with significant light pollution. The glow of building lights that surround the telescope could prevent a star's visibility and make it difficult to render the correct aperture and annulus for analysis.

5 Conclusion

This report concludes that RU Cam is a low magnitude Type II Cepheid variable star. The original goal was to determine the apparent magnitude and flux density of this celestial body. It was achieved within a 20% margin of error. In the event I repeat this experiment, I would take all of my measurements on the same date to reduce any variability between the star and my calibration frames. Flux density is important to measure because it allows scientists to determine characteristics of the star like distance, temperature, and luminosity.

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References

[1] W. Ceraski. "Deux nouvelles variables". In: *Astronomische Nachrichten* 166 (Jan. 1904), pp. 173–174. DOI: 10.1002/asna.19041661108. URL: https://zenodo.org/records/1424841 (visited on 02/16/2026).

[2] Susmita Das, Anupam Bhardwaj, and Marcella Marconi. "Type II Cepheids: period-luminosity-metallicity relations for the Population II distance scale". In: *Frontiers in Astronomy and Space Sciences* 12 (Jan. 2026). DOI: 10.3389/fspas.2025.1718800. (Visited on 02/17/2026).

[3] Tonu Kipper and Valentina Klochkova. "Optical spectroscopy of RU Cam, a pulsating carbon star". In: *Baltic Astronomy* 16 (June 2007), pp. 383–396. DOI: 10.48550/arxiv.0706.2969. URL: https://ui.adsabs.harvard.edu/abs/2007BaltA..16..383K/abstract (visited on 02/16/2026).

[4] Z. Kollath and B Szeidl. "On the irregular light variation of RU Camelopardalis." In: *Astronomy and Astrophysics* 277 (Sept. 1993), pp. 62–68. URL: https://ui.adsabs.harvard.edu/abs/1993A%26A...277...62K/abstract (visited on 02/16/2026).

[5] Henrietta Leavitt. "1777 Variables in the Magellanic Clouds". In: *Annals of Harvard College Observatory* 60 (1908). URL: https://articles.adsabs.harvard.edu//full/1907AnHar..60...87L/0000100.000.html (visited on 02/16/2026).

[6] Henrietta S. Leavitt and Edward C. Pickering. "Periods of 25 Variable Stars in the Small Magellanic Cloud." In: *Harvard College Observatory Circular* 173 (Mar. 1912), pp. 1–3. URL: https://ui.adsabs.harvard.edu/abs/1912HarCi.173...1L/abstract.

[7] George Wallerstein. "The Cepheids of Population II and Related Stars". In: *Publications of the Astronomical Society of the Pacific* 114 (July 2002), pp. 689–699. DOI: 10.1086/341698. URL: https://ui.adsabs.harvard.edu/abs/2002PASP..114..689W/abstract (visited on 02/16/2026).

[8] M. Wenger et al. "The SIMBAD astronomical database. The CDS reference database for astronomical objects". In: *Astronomy and Astrophysics Supplement Series* 143 (Apr. 2000), pp. 9–22. DOI: 10.1051/aas:2000332.

[9] G.V. Zaitseva. "Cycles of Activity in RU Camelopardalis and an Evolutionary Interpretation of Its Photometric Behavior". In: *Soviet Astronomy* 16 (Apr. 1973), p. 856. URL: https://ui.adsabs.harvard.edu/abs/1973SvA....16..856Z/abstract (visited on 02/17/2026).